

CONCISE DESCRIPTION OF RELEVANCE: THIRD-PARTY SUBMISSION UNDER 37 CFR § 1.290

U.S. Application No. 19/318,450 (Publication No. US 2026/0070232 A1)

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Title: permanent magnet chuck

This document discloses a permanent magnet chuck that effectively magnetizes the whole magnetic zone of a surface plate and improves its magnetic attractive force by switching the surface plate between magnetized and non-magnetized conditions. Its potential relevance to the claims of the above-identified application is set forth below.

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 1: “a magnet which is arranged in the housing and which is displaceable along a displacement direction between a gripping position for gripping the ferromagnetic workpiece and a release position for releasing the ferromagnetic workpiece”	The reference discloses a permanent magnet able to move in the vertical direction between a bottom plate and the surface plate, which corresponds to a magnet displaceable along a displacement direction between a gripping and release position. “a permanent magnet able to move in the vertical direction between a bottom plate and the surface plate.” • Abstract
Claim 1: “a number of pole shoes attached to the housing, wherein each pole shoe has a workpiece contact surface and is designed to guide a magnetic field portion of the magnet to the workpiece contact surface”	The reference discloses a magnetic pole part 14 brought into contact with a yoke 26, and each magnetic zone 14a of a surface plate, brought into contact with a yoke upper magnetic part 24b, which are alternately magnetized and guide magnetic flux. “a magnetic pole part 14, brought into contact with a yoke 26, and each magnetic zone 14a of a surface plate, brought into contact with a yoke upper magnetic part 24b, are alternately magnetized in a different magnetic pole placing ...” • Abstract
Claim 2 (the displacement direction is aligned parallel to the longitudinal axis of the housing)	The reference discloses permanent magnets 20 that are rigidly supported by each frame 34 so as to be aligned in the same magnetization direction, and move in the vertical direction. “The permanent magnets 20 are rigidly supported by each frame 34 so as to be aligned in the same magnetization direction, and as shown in FIG. 26, that is, the magnetic a1 of the face plate 16? The magnetization ...” • Paragraph [0016]

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 3 (The magnetic gripper according to claim 1, wherein each pole shoe has an active structure for directing the magnetic field portion)	The reference discloses a magnetic pole part 14 brought into contact with a yoke 26, and each magnetic zone 14a of a surface plate, brought into contact with a yoke, which direct the magnetic field. “In the upper position of a frame structure 34 supporting a magnet 20, a magnetic pole part 14, brought into contact with a yoke 26, and each magnetic zone 14a of a surface plate, brought into contact with a yoke ...” • Abstract
Claim 6 (the active structure has a recess between the first projection and the second projection)	The reference discloses a recess 32 for the permanent magnet 20 defined between the adjacent magnetic yokes 24 and 26 by spacers 28 and 30. “A recess 32 for the permanent magnet 20 is defined between the adjacent magnetic yokes 24 and 26 by spacers 28 and 30 in which an upper spacer 28 and a lower spacer 30 made of a non-magnetic material are ...” • Paragraph [0013]
Claim 8 (the magnet is arranged in the housing such that, in the gripping position, the first pole shoe acts as a north pole and the second pole shoe acts as a south pole)	The reference discloses that magnetic pole parts are alternately magnetized in a different magnetic pole, placing the surface plate 16 in a magnetized condition because a magnetic pole surface adapted to a yoke 24 of each permanent magnet 20 is brought into contact with only the upper part. “are alternately magnetized in a different magnetic pole placing the surface plate 16 in a magnetized condition because a magnetic pole surface adapted to a yoke 24 of each permanent magnet 20 is brought into contact with only the upper ...” • Abstract
Claim 14 (wherein, in a release state, the magnet is in the release position and the ferromagnetic workpiece is not pressed, by means of a magnetic force, against each workpiece contact ...)	The reference discloses that in the bottom position of the frame structure 34, the surface plate 16 is placed in a non-magnetized condition by forming closed loop magnetic flux via the yoke 26, yoke bottom magnetic part 24C and a bottom plate 18, meaning the ferromagnetic workpiece is not pressed by magnetic force. “Next in the bottom position of the frame structure 34, the surface plate 16 is placed in a non-magnetized condition by forming closed loop magnetic flux via the yoke 26, yoke bottom magnetic part 24C and a bottom plate 18 ...” • Abstract • Claim 24

Claim language (US 2026/0070232 A1)	Relevant disclosure of this document
Claim 16 (The magnetic gripper according to claim 1, wherein the receptacle is in the form of a groove)	The reference discloses a recess 32 for the permanent magnet 20 defined between the adjacent magnetic yokes 24 and 26. “A recess 32 for the permanent magnet 20 is defined between the adjacent magnetic yokes 24 and 26 by spacers 28 and 30 in which an upper spacer 28 and a lower spacer 30 made of a non-magnetic material are ...” • Paragraph [0013]